

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10 Total Marks: 50 Annexure A

Code of Conduct:

I KNeeha (19242245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the student:K.Neeha

Scenario-Based Assessment DURATION: 1.5 Hrs 1. Smart Traffic Signal Control

Using Reinforcement Learning

(5 Marks)

A smart-city transportation system wants to reduce traffic congestion by automatically controlling traffic signals at busy road intersections. The system plans to use Reinforcement Learning (RL) to learn traffic patterns from vehicle density, waiting time, and traffic flow. The RL agent observes the current traffic condition and selects suitable signal-control actions to improve the movement of vehicles.

Report how the RL framework can be applied to this scenario by defining the possible states, actions, and reward functions. Explain how the agent can learn an effective traffic-signal control policy through repeated interaction with the traffic environment. Represent the complete RL decision-making process using a suitable flowchart.

Suggested RL Components

RL Component	Possible Representation
Agent	Intelligent traffic signal controller
Environment	Road intersection and traffic flow
States	Vehicle density, queue length, waiting time, signal status
Actions	Keep signal, change to green, change to red, change signal phase

Positive Reward Reduced waiting time and smooth traffic flow

Negative Reward Traffic congestion, long waiting time, unnecessary signal changes

Policy Strategy for selecting the appropriate signal action

Flowchart

Traffic Data → Identify State → RL Agent → Select Signal Action → Traffic Response → Calculate Reward → Update Policy → Improved Signal Control

2. Autonomous Delivery Drone – Exploration vs Exploitation

(5 Marks)

A logistics company uses an autonomous delivery drone to transport packages between a warehouse and different customer locations. The drone uses Reinforcement Learning to learn efficient flight paths while avoiding buildings, restricted areas, obstacles, and changing weather conditions. During training, the drone repeatedly selects a familiar route because it

provides a high reward, even though alternative routes may be shorter, safer, or more energy-efficient.

Analyze how the exploration vs exploitation trade-off affects the drone's navigation performance. Explain how insufficient exploration can cause the drone to select suboptimal routes and suggest suitable improvements, such as ϵ -greedy action selection, epsilon decay, and reward-based route optimization, to achieve a better balance between exploring new routes and exploiting previously learned routes. Represent the learning process using a suitable flowchart.

Exploration vs Exploitation

Aspect	Exploration	Exploitation
Benefit	Trying new routes/actions	Selecting the best-known route
	Purpose Discover better alternatives	Use existing knowledge
	May find shorter/safer routes	
		Provides reliable decisions
Risk	Trying an unfamiliar flight path	May choose poor routes May remain stuck with a longer route Reusing a successful route

Possible Drone RL Elements

Component	Example
State	Drone position, battery level, obstacle location, weather
Action	Move forward, turn left/right, change altitude, select another route
Reward	Shorter route, successful delivery, obstacle avoidance
Penalty	Collision, unnecessary movement, excessive energy consumption
Exploration Methodology	Gradually decrease ϵ during training
Improvement	

Effect of Insufficient Exploration

Known Route $\rightarrow$ High Reward $\rightarrow$ Repeated Selection $\rightarrow$ Alternative Routes Not Explored $\rightarrow$ Better Route Not Discovered $\rightarrow$ Longer Navigation

Flowchart

Observe Drone State $\rightarrow$ Exploration/Exploitation Decision $\rightarrow$ Select Route $\rightarrow$ Execute Action $\rightarrow$ Receive Reward/Penalty $\rightarrow$ Update Policy $\rightarrow$ Adjust ϵ $\rightarrow$ Repeat Training

Expected Focus

The answer should explain that too much exploitation can make the drone repeatedly follow a suboptimal route, while too much exploration can cause unnecessary route changes. A suitable balance allows the drone to discover better routes during training and gradually exploit the most efficient learned route.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Understandin g of Scenario	20%	Demonstrat es clear and complete understandi n g of the scenario and context	Shows good understandi n g with minor gaps	Partial understandin g ; misses key aspects	Misinterpr et s or fails to understand the scenario
Identificati on of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided